307307 BI Methods

Introduction to Agentic AI

https://www.knime.com/events/data-aware-agentic-ai-getting-started-course

Agentic AI

- Agentic AI is an **AI system that can plan, decide, and act** to achieve goals, rather than just respond passively to prompts.
- It combines LLMs with tools, memory, and reasoning capabilities to perform tasks autonomously.

Key Characteristics

- **Goal-driven**: Works toward objectives instead of isolated answers.
- **Tool-using**: Can call external functions, databases, or APIs to gather information or take action.
- **Memory-aware**: Retains past interactions and context across steps.
- **Reasoning loop**: Plans multi-step actions, executes them, evaluates results, and adapts.

Why it matters

- Moves beyond Q&A into **problem-solving agents**.
- Enables automation of workflows, research, customer support, and decision-making.
- Foundation for building **personal AI assistants** and **autonomous applications**.

Agentic AI Scenario - Travel Assistant

Task: Arrange a budget-friendly trip.

Steps the Agent Takes

1. **Understand the goal**
    - User: *"Find me the cheapest way to travel to Paris next weekend, plus affordable lodging and a rental car."*
2. **Plan the workflow**
    - Search for flights and compare fares.
    - Find nearby hotels with availability and within budget.
    - Check for rental cars near the hotel.
    - Combine results into a single travel plan.
3. **Use external tools**
    - Flight search API → retrieves cheapest available flight.
    - Hotel booking API → filters for price, location, and vacancy.
    - Car rental service API → checks for affordable, available options.
4. **Evaluate and optimize**
    - Ensures hotel is close to airport or city center.
    - Chooses options that minimize total cost.
    - Cross-checks dates and availability.
5. **Report and act**
    - Returns the best itinerary to the user.
    - Optionally books the flight, hotel, and car.
    - Provides confirmation details in one place.

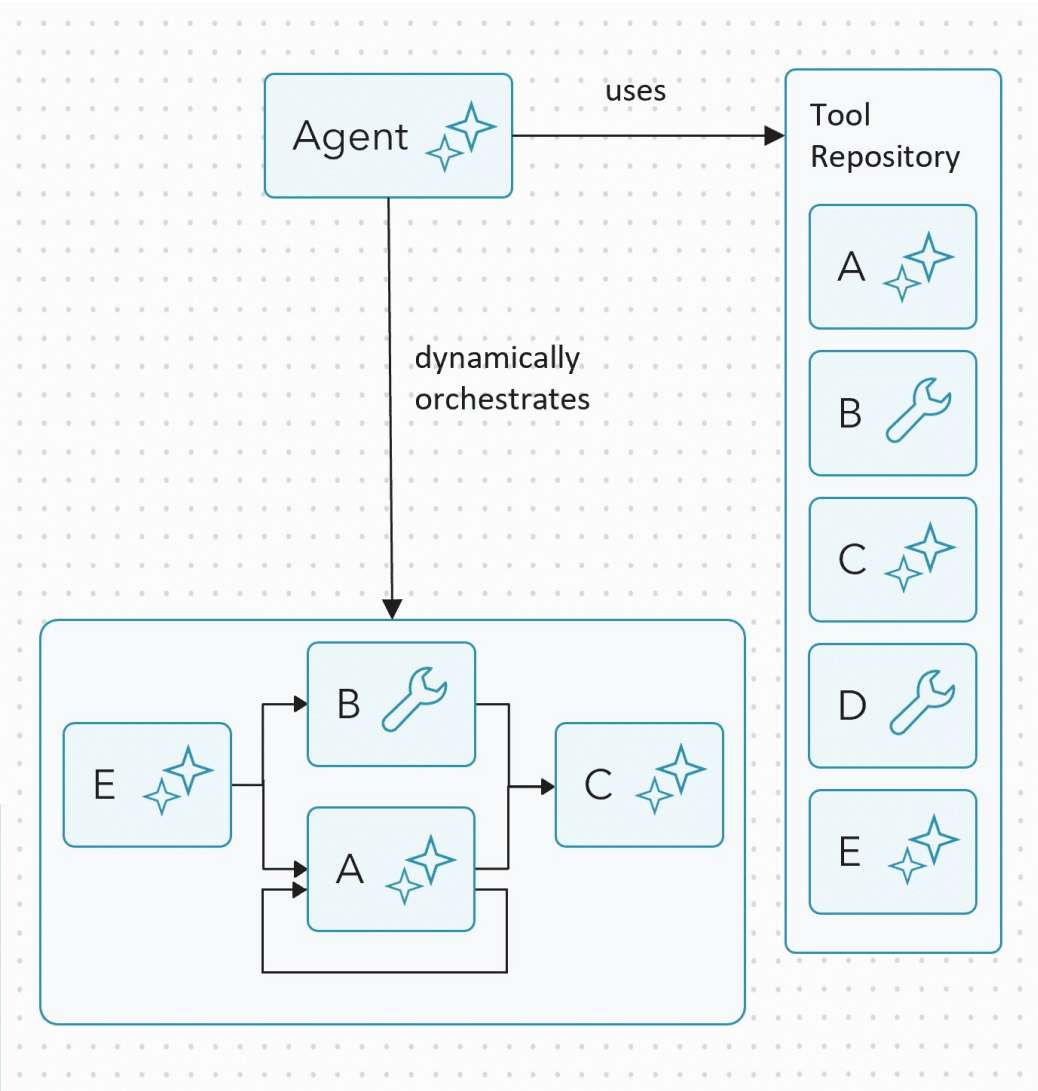
Outcome: The agent delivers a **complete, optimized travel plan** instead of just answering one query.

Building Blocks of Agentic AI

- Agentic AI consists of a collection of tools specialized in different tasks, and an agent that calls them to accomplish the request by the user.
- The agent accesses the tools from a repository, combining them dynamically to accomplish the required process.
- Agentic AI accomplishes the previous scenario by combining and orchestrating different tools to complete a required task.
- Based on the input from the user, an AI agent identifies necessary tools, combines them in a correct order, monitors their executions, mediates their inputs and outputs, and formulates the final outcome the user can understand.

What is an Agent?

- An agent is an autonomous software system that can perceive its environment, reasoning about it, making decisions, and taking actions to achieve specific goals.
- It can combine tools at its disposal to achieve the goal to produce the desired outcome.



Tools in KNIME

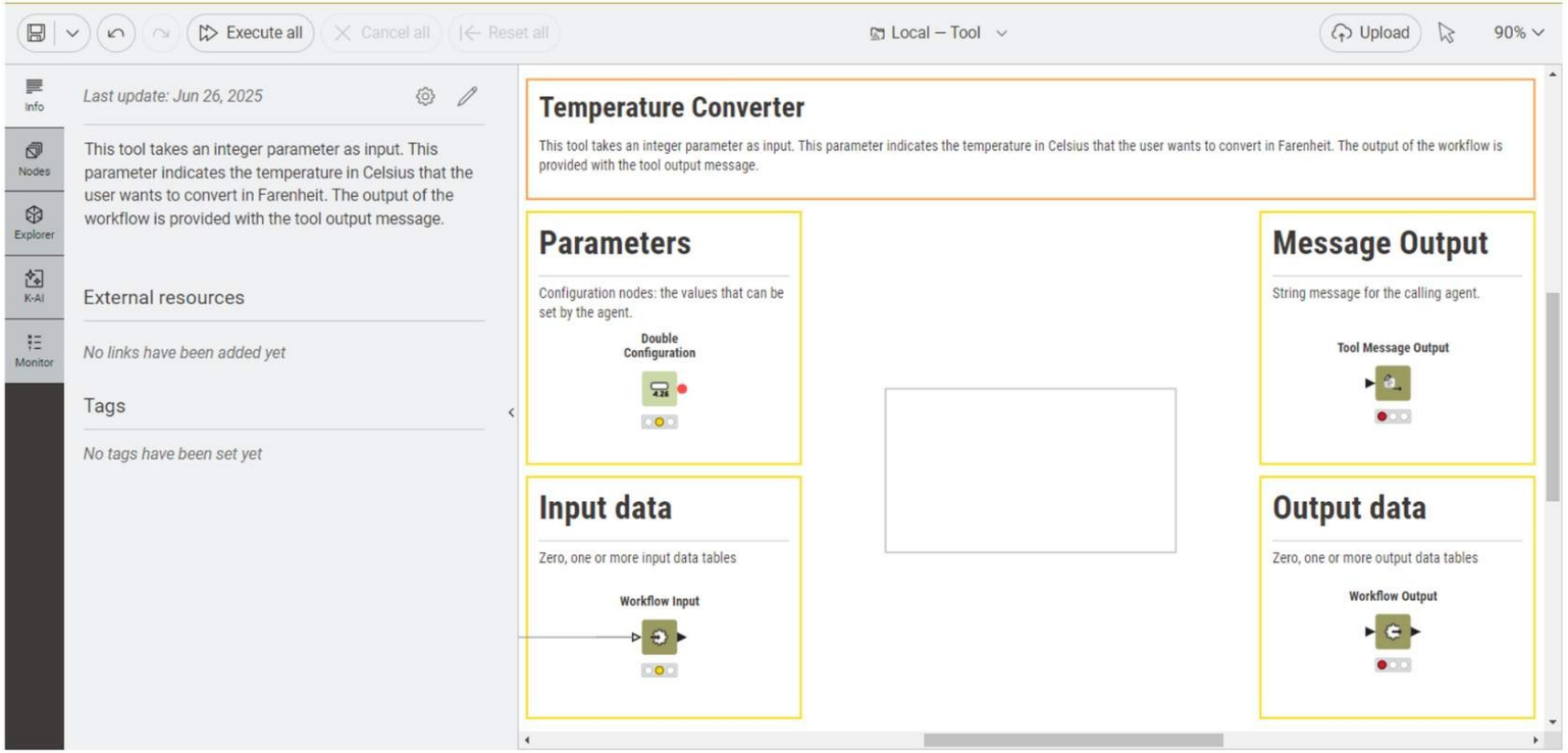
- Tools in KNIME are workflows specialized in different tasks.
- Such tools can be a modified version of an existing workflow, or constructed from scratch.
- Tools can optionally receive data input from an agent or return a data output to an agent.
- We have two classes of tools: classic tools and intelligent tools:
  - Classic Tools
    - Perform a designated task without using AI
    - Examples: data aggregation, classification, numeric prediction, data transformation
  - Intelligent Tools
    - Perform a designated task using AI models
    - Examples: summarizing document, machine translation, sentiment analysis





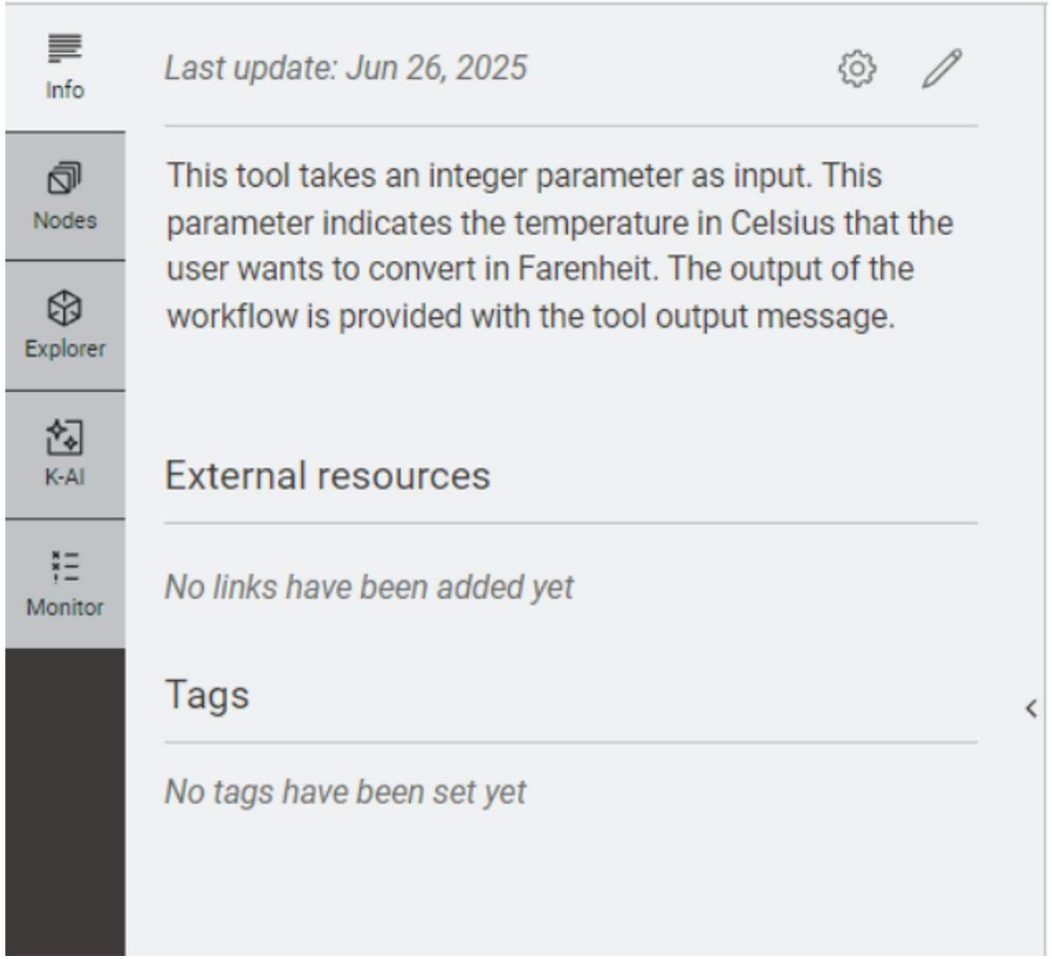
Tools in KNIME

- A tool in KNIME is a KNIME workflow that performs a specific task.
- However, the tool workflow in KNIME needs to contain certain elements that are required to make the workflow a Tool usable by the agent



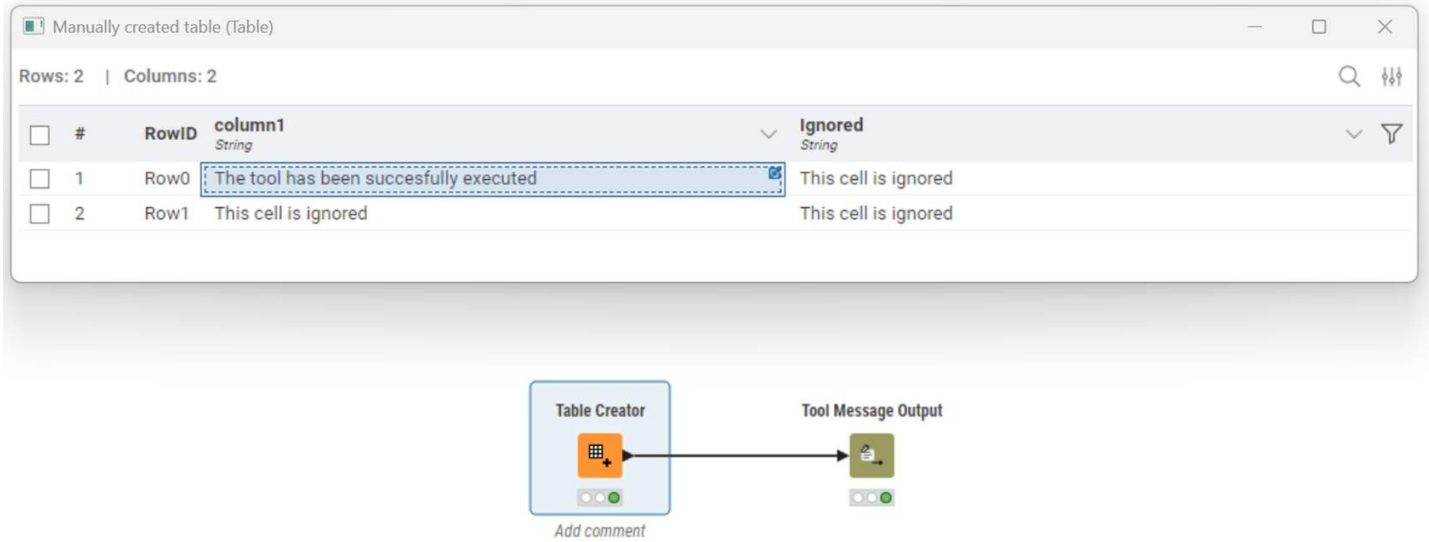
1-Tool Description

- Tool description explains as precisely as possible what task the tool is able to solve.
- It is defined in the workflow's Info field.
- The agent reads this description and decides when to use the tool.
- A well-written description allows the agent to reason effectively about the available options.
- We can also include some examples of usage, to enhance the description.
- The description should explain:
  - The task performed by the tool
  - The expected input data
  - The output produced
  - The configurations required
  - The types of question the tool is designed to answer



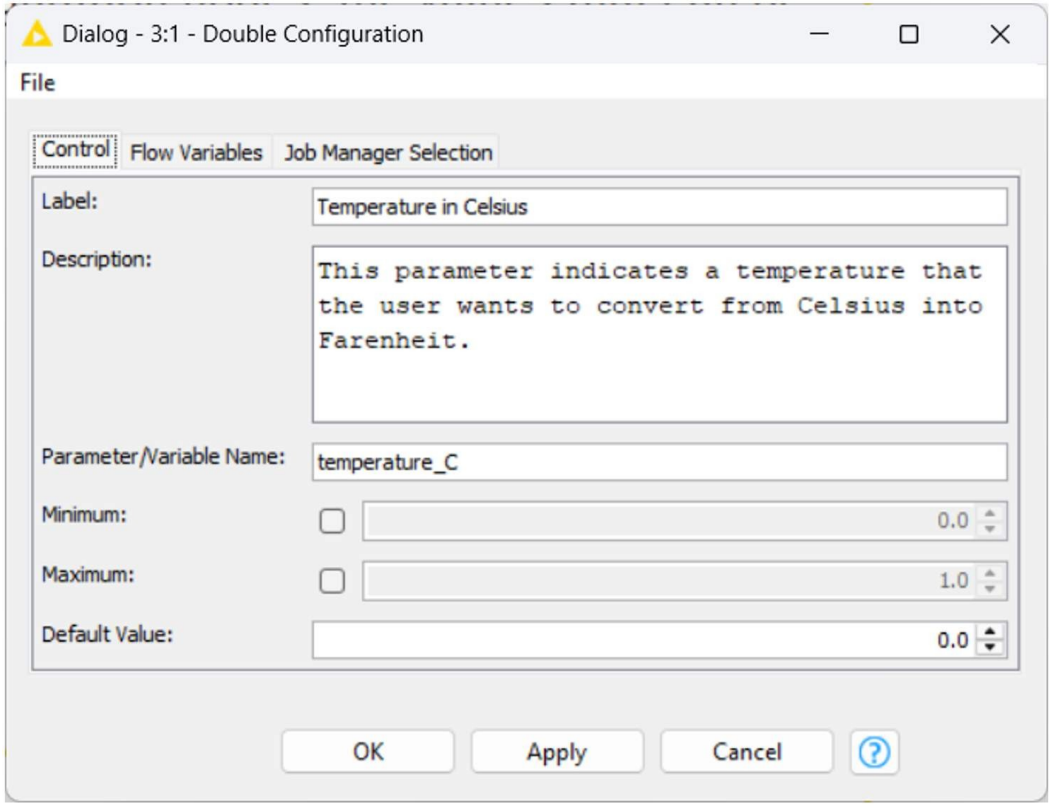
2-Tool Message Output

- The Tool Message Output node provides optional feedback to the agent after a tool execution.
- We can include this node if textual output is needed for the agent to reason with after the tool call (remove it if no output is required from the tool).
- The node reads the first value from the first cell of its input table.
- This string becomes the content of the Tool Message returned to the agent. It is useful, for example, to return
  - A summary of the processed data (“The database contains data about 22 customers.”)
  - Short textual insights (“The average temperature is 22,5°C which is equivalent to 72.5°F.”)
  - Confirmation or intermediate results (“The email has been successfully sent to the recipient list.”)



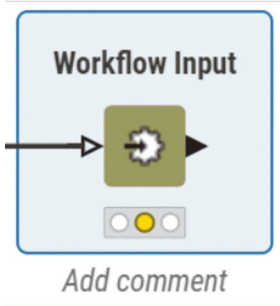
3-Parameters

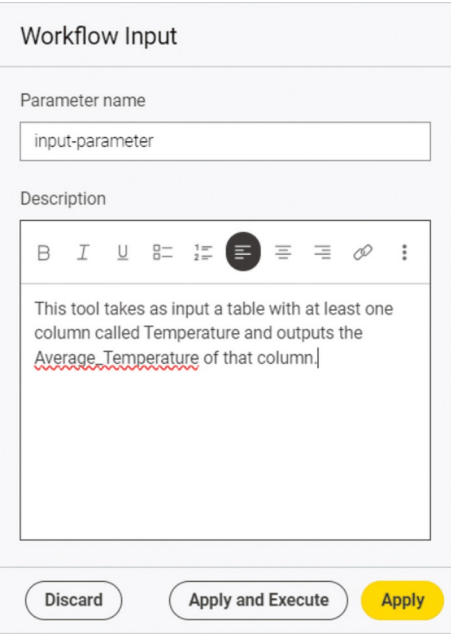
- Parameters are values that can be set by the agent.
- We can use configuration nodes (such as String Configuration or Integer Configuration) to define adjustable parameters.
- For each parameter we need to:
  - Provide a clear parameter name used as the variable name
  - Write a concise description explaining its purpose.
- The agent reads these definitions to determine which parameter values to set during tool execution

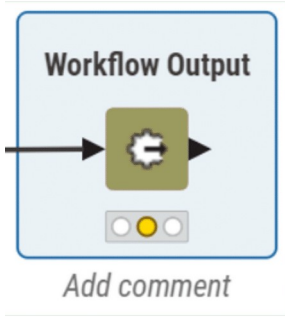


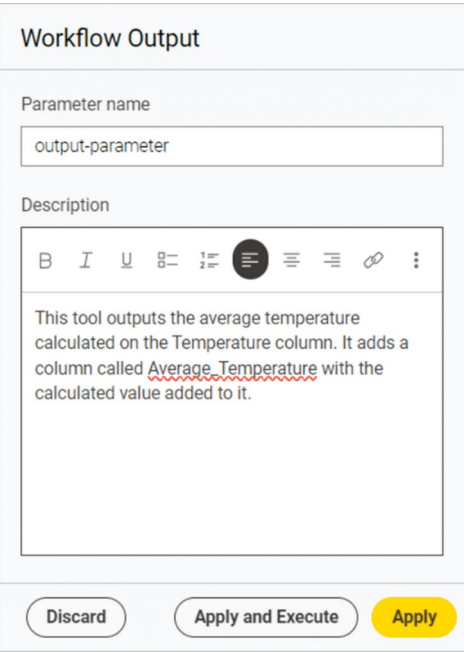
4-Data Input & Data Output (optional)

- Tools can optionally take input data or send a data output or both.
- Use the Workflow Input node to define the incoming data structure and the Workflow Output node to specify the result table the tool produces.
- The agent does not access raw data directly but can trigger tools that process and summarize data as needed.









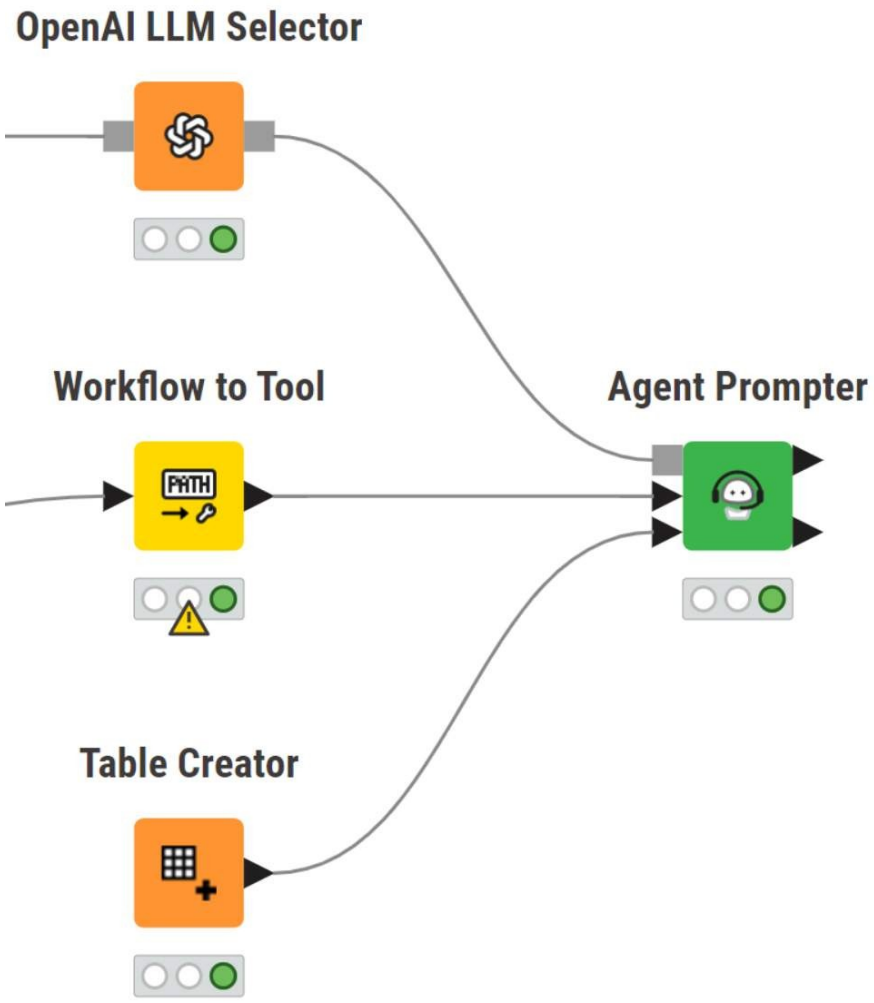
How does Agents Work?

- AI Agents work by a series of reasoning loop executions.
- In a way, agents are performing trials and errors until the required task is completed.
- Here are the steps in agents' execution:
  - The agent thinks about the task.
  - Chooses whether to call a tool.
  - Evaluates the result produced by the tool.
  - Decides if further steps are needed.
  - Continues until the task is completed.



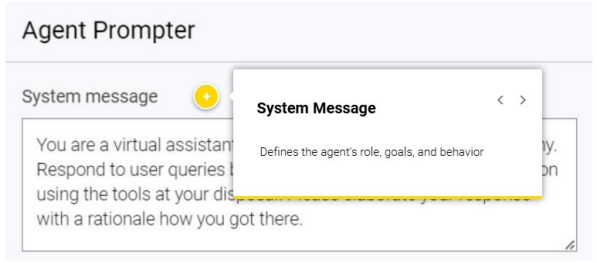
Agent Prompter

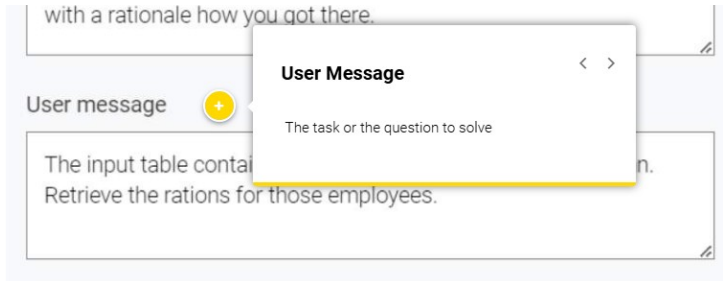
- In KNIME Analytics Platform, an AI agent is implemented by the Agent Prompter node, executing the loop described above.
- Agent Prompter Inputs:
  1. A connection to the selected LLM Model
  2. List of tools available to the agent
  3. Optional input data if some tools require external data.
- Agent Prompter Outputs:
  1. Result of execution showing interaction between agent and tools.
  2. Optional output data.



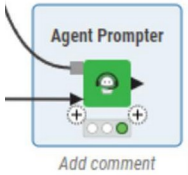
Agent Prompter Configuration Panel

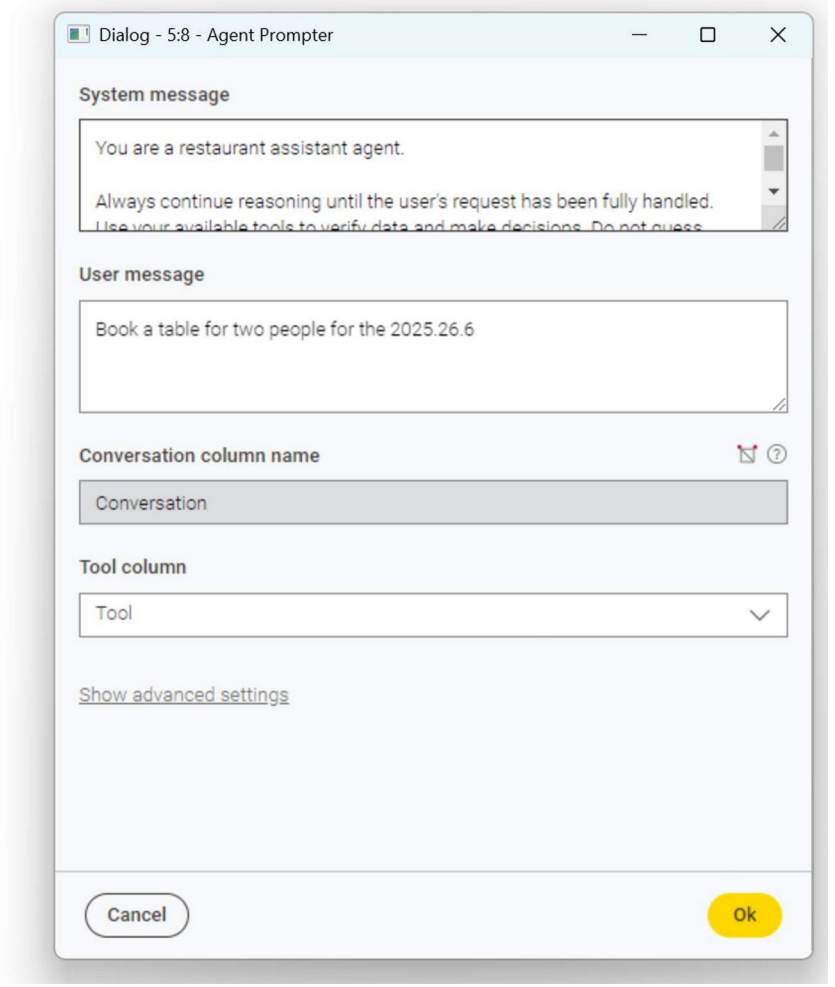
In the configuration panel, we supply the information necessary to execute the Agent Prompter node





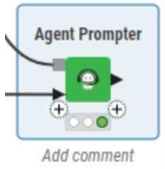






Example

The Agent Prompter then outputs the conversation between the user, the agent, and the tool.



Conversation (Table)

Rows: 5 | Columns: 1

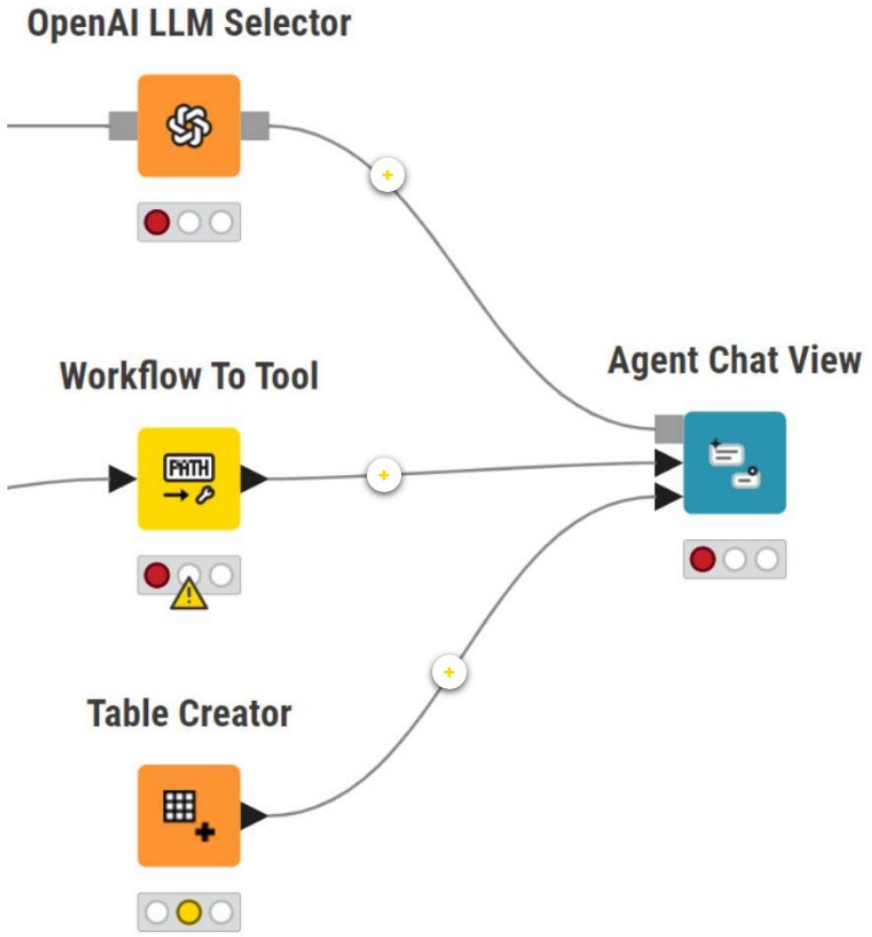
#	RowID	Conversation *Message*
1	Row0	User **Data Tools Interface** You have access to tools that can consume and produce data. The interaction with these tools is mediated via a data repository that keeps track of all available data items. The repository is represented as a map from IDs to data items
2	Row1	User Book a table for two people for the 2025.26.6

3	Row2	AI
		table_availabilty
		`configuration:`
		`  booking_date-2: "2025-06-26T00:00:00+02:00[Europe/Berlin]"`
		`  number_people-4: 2`

4	Row3	table_availabilty The booking for table T1 with 2 people, on 2025-06-26 was confirmed!
5	Row4	AI The table for two people has been successfully booked for June 26, 2025.

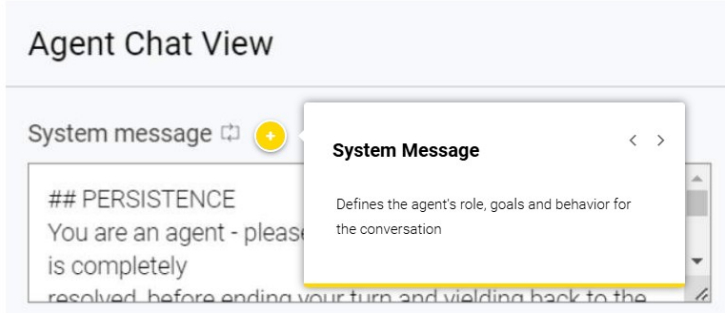
the Agent Chat View Node

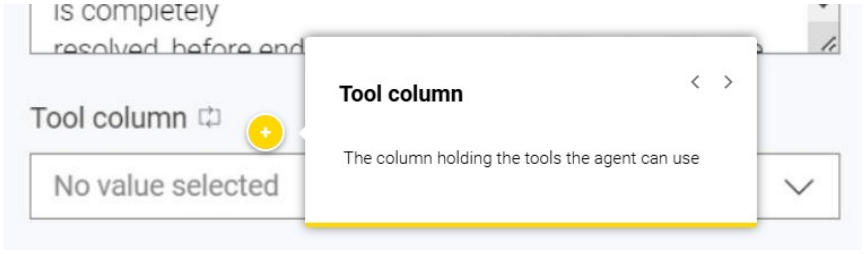
While the Agent Prompter node takes a single prompt and returns the result, the Agent Chat View node provides an interactive chat interface for live agent conversations.

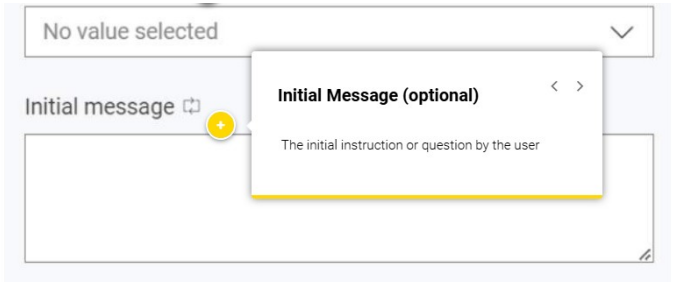


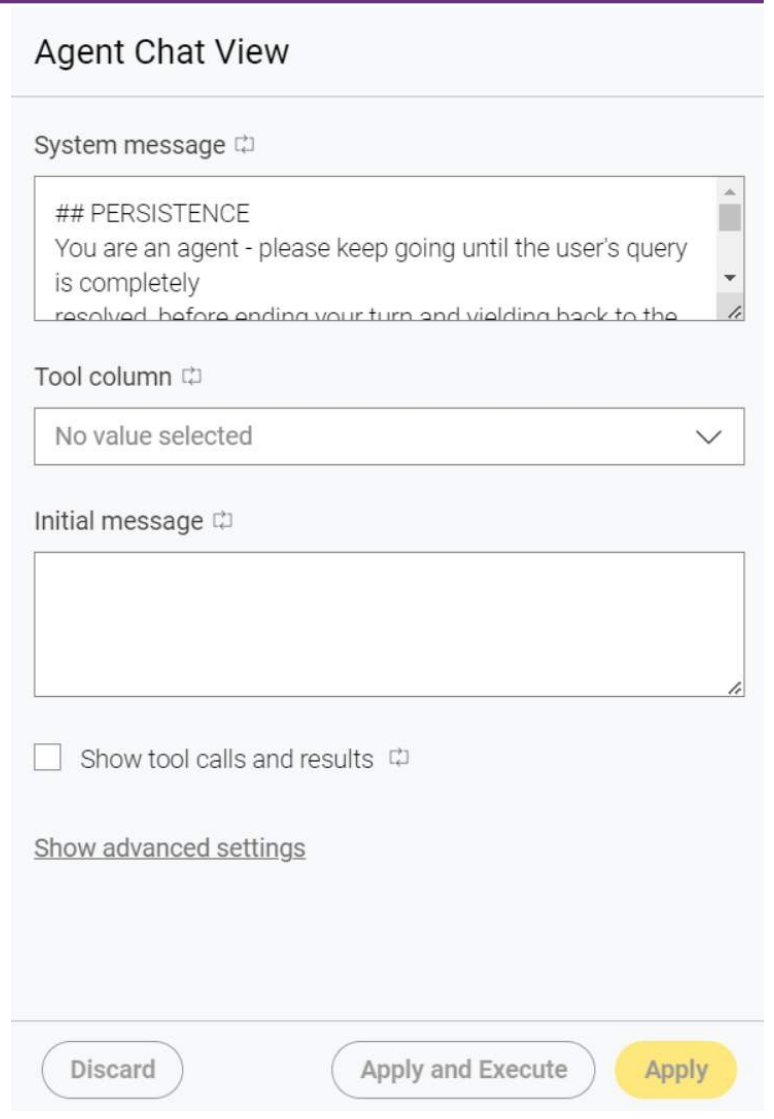
Agent Chat View Configuration Panel

We supply the information necessary to execute the Agent Chat View node in the c





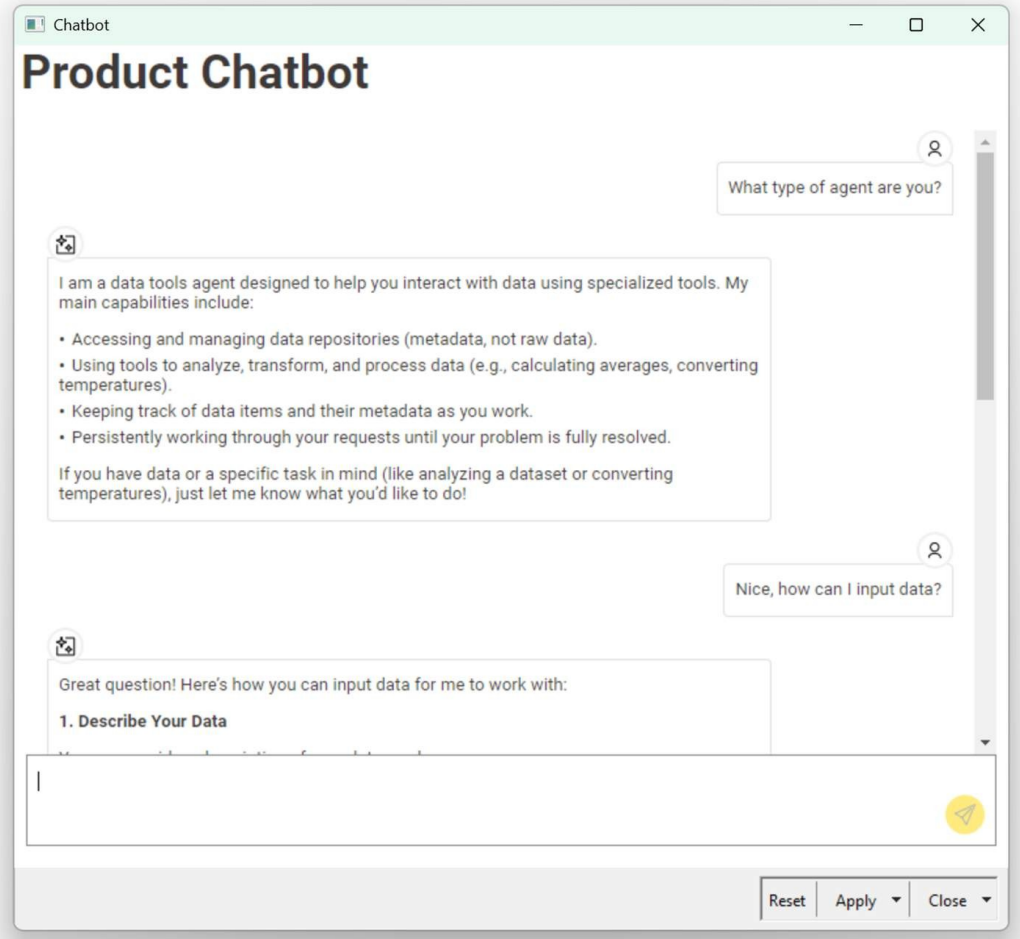


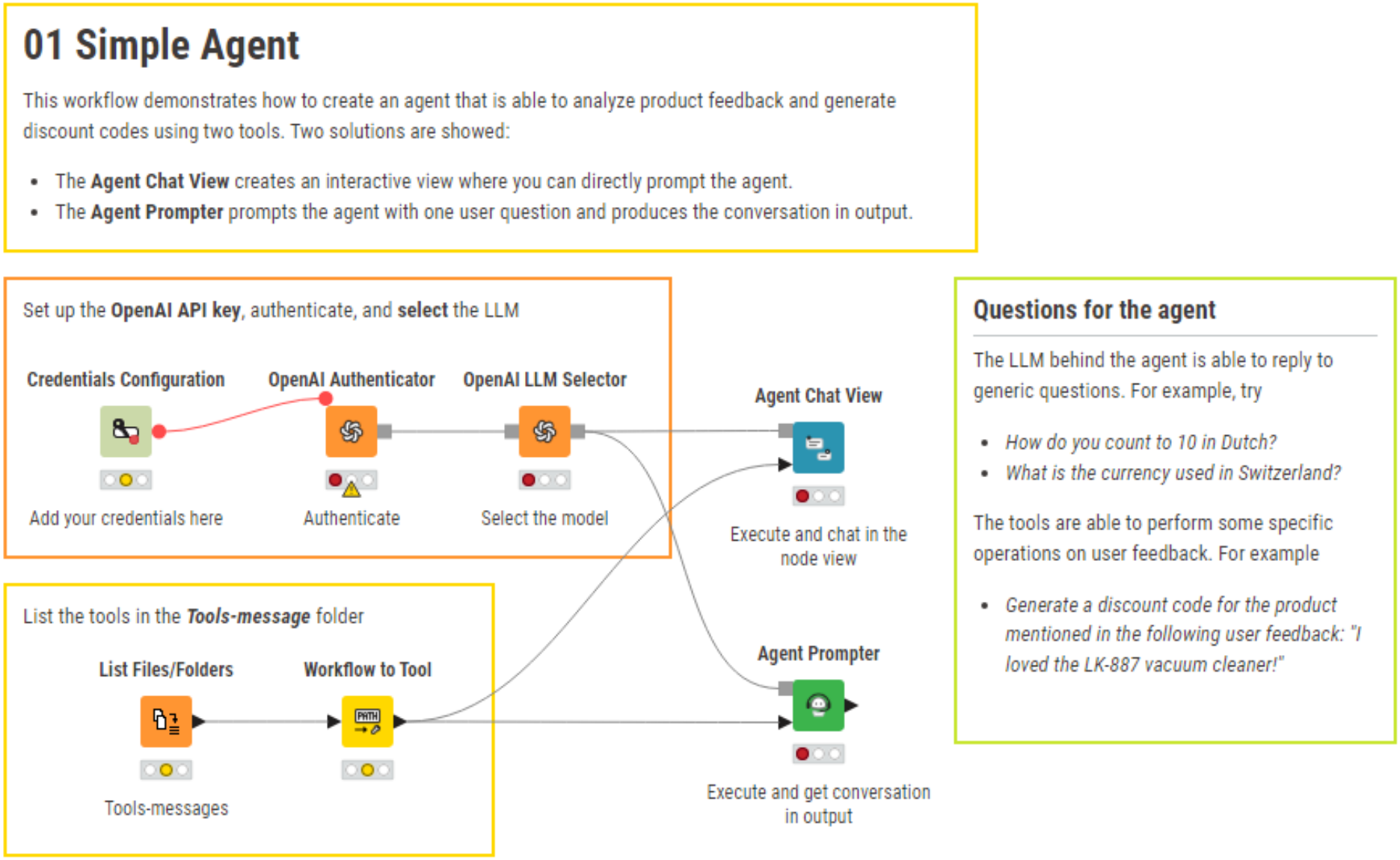


Example

- During execution, the node opens a chat interface where additional user inputs can be provided interactively.
- The agent reasons, selects tools, processes results, and generates responses in real time.
- This node can be embedded inside a component, allowing it to be deployed and shared via KNIME Business Hub, making the agent available as an interactive application to end users.







This workflow can be downloaded as following:

1. Download Course Workflows from VClass
2. Goto Generative AI Folder -> Getting Started Courses -> Data Aware Agentic AI – Getting Started
3. Open 01 Simple Agent

Example Workflows

KNIME Learning Center

Download the example workflows

This course does not include exercises. To earn your microcredentials, you'll need to complete a knowledge check at the end.

You can, however, download the example workflows featured throughout the course and explore them to apply the knowledge you have acquired.

Download the **example workflows** from the KNIME Community Hub.

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